CS170: Introduction to Artificial Intelligence, Dr. Eamonn Keogh

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Project for CS 170 Winter 2026, with Dr. Eamonn Keogh

In completing this assignment, I consulted:
- The project 2 sample report from this class
- Project 2 readme, briefing, hints, cheat sheet from the professor
- Python 3 standard library: https://docs.python.org/3/library/index.html
- Numpy documentation/user guide: https://numpy.org/doc/stable/user/index.html#user
- Slides and videos provided by the professor
- Sanity Check Data Sets 1 and 2

In completing this assignment, for time-saving purposes, I used two different laptops with different OS and specifications to run the traces: Windows OS and MacOS

All important code is original, except:
1) I converted the given pseudo code from the professor's briefing video and cheat sheet to **Python** for the forward search
2) I used the functions from **NumPy** and **Time** for data management and execution times.

Outline of this report:

- Cover Page (This page)
- My report: Pages 2 to 7
- Sample trace on forward selection with the small data set, pages 8 to 11
- Sample trace on backwards elimination with the small data set, pages 12 to 15
- My code is available in the GitHub repository linked on page 15 and this in this URL: https://github.com/kayliezhao/cs170_lab_2

CS170: Project 2: Feature Selection with Nearest Neighbor

Kaylie Zhao Mar-16-2026

Introduction

The nearest neighbor classifier is a classifier that can be used to identify what class an object may belong to, essentially solving the classification problem. According to Dr. Eamonn Keogh, the nearest neighbor algorithm is "sensitive to the units of measurement." This means that because it relies on measuring the distance between data points, features with larger scales can overpower the smaller ones.

In this project, we are tasked with implementing the nearest neighbor classifier to use it within a "wrapper" to perform Feature Selection (Keogh). The goal is to take 2 different datasets, one large and one small, to find the smallest subset of the features with the highest classification accuracy. Two searches were used to implement this classifier, forward selection and backward elimination.

This project will also show sample traces of forward selection and backward elimination on the small data set assigned to me, CS170_Small_DataSet__85.txt. The technology stack used for this project was Python3 for coding, NumPy library for data management, Time library for execution times, GitHub for the repository, VSCode for the IDE, and Google Sheets for the visualization of my data in the form of bar charts.

Forward Selection on the Small Dataset

In Figure 1, we see the result of running forward selection on the dataset assigned to me, CS170_Small_DataSet__85.txt.

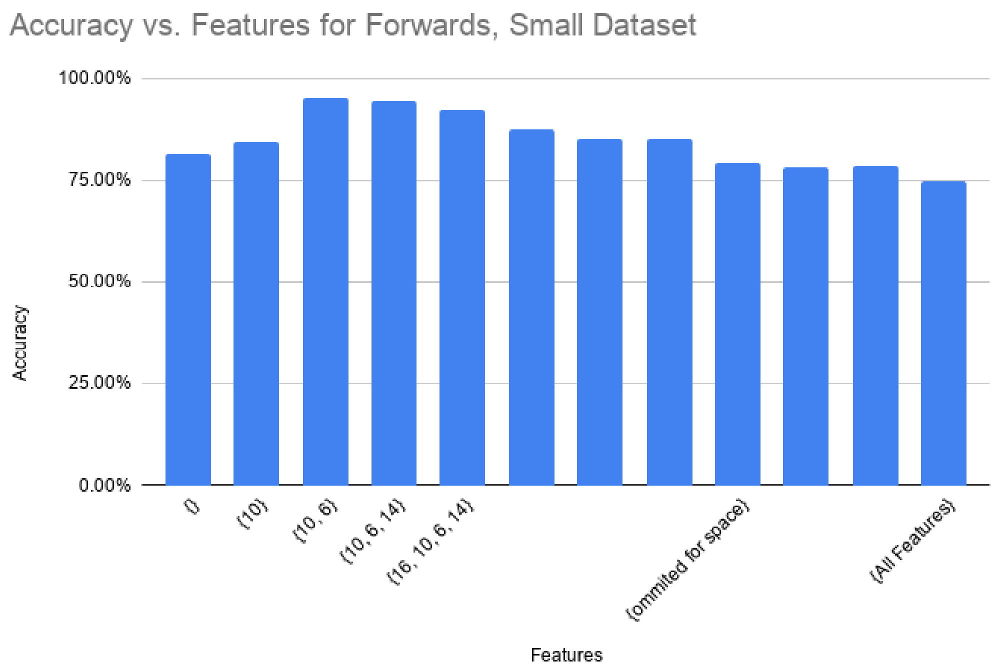

Figure 1: Accuracy of subsets of features discovered by forward selection on the small dataset. Features between the 4 features and all features were omitted to save space.

As shown in Figure 1, we have no features at the beginning of our search, so I reported the default rate of 81.40% for the small dataset. The default rate is “the size of the most common class, over the size of the full dataset” (Keogh). By adding feature ‘10’ to the subset, the accuracy increased by 3.2% to become 84.6% accuracy. Adding feature ‘6’ to the subset increased the accuracy to become the best accuracy of 95.2%, increasing by 10.6%. However, each additional feature added after the ‘6’ gradually decreased in accuracy until we reached the end with all features with an accuracy of 74.60%.

Backward Elimination on the Small Dataset

Next, as shown in Figure 2, we see the result of running backward elimination on the dataset assigned to me, CS170_Small_DataSet__85.txt.

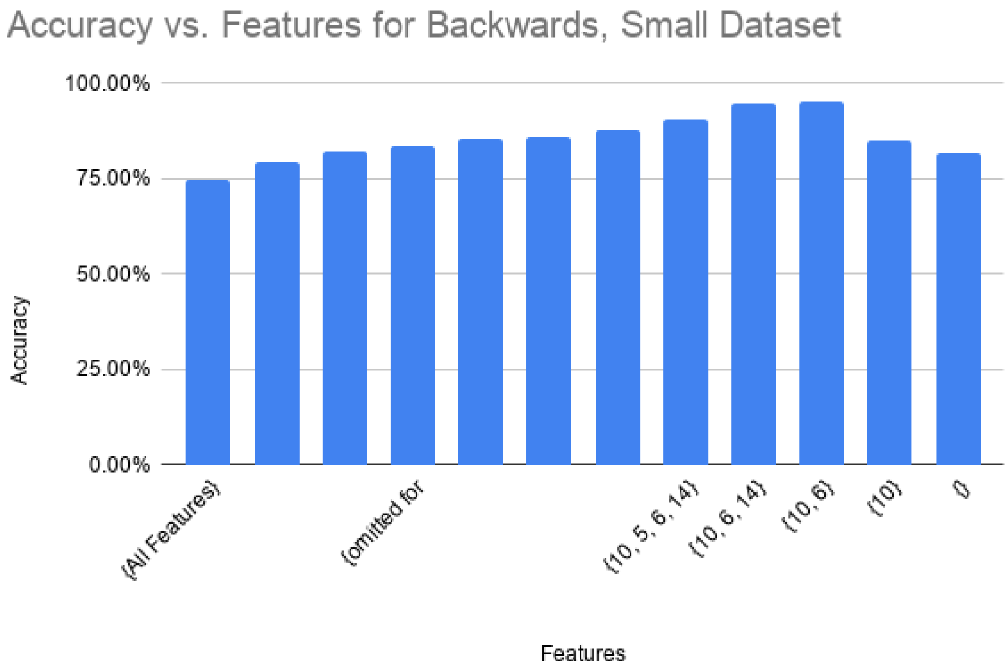

Figure 2: Accuracy of subsets of features discovered by backward elimination on the small dataset. Features between all features and 4 features were omitted to save space.

As shown in the backward elimination bar chart in Figure 2, the last tested subset in forward selection has the same accuracy as the first tested subset with all features in backward elimination. The accuracy of 74.60% confirms that my evaluation was consistent and I could proceed with accurate data. In addition to this, a sanity check was performed using the given first sanity check data set, and the percentages also matched up.

Backward elimination for the small dataset showed the same best accuracy for subset features {10, 6} at accuracy 95.20%. It showed a gradual increase in accuracy until we hit the peak of features 10 and 6, then decreased until the empty set.

Conclusion for the Small Dataset

Overall, based on Figures 1 and 2 that showcase forward selection and backward elimination on the small dataset, we can conclude that the best accuracy of the searches was features {10, 6} with an accuracy of 95.2%.

Forward Selection on the Large Dataset

Now, in Figure 3, I will be switching to a larger dataset with 64 features and 3000 instances assigned to me, CS170_Large_DataSet__98.txt.

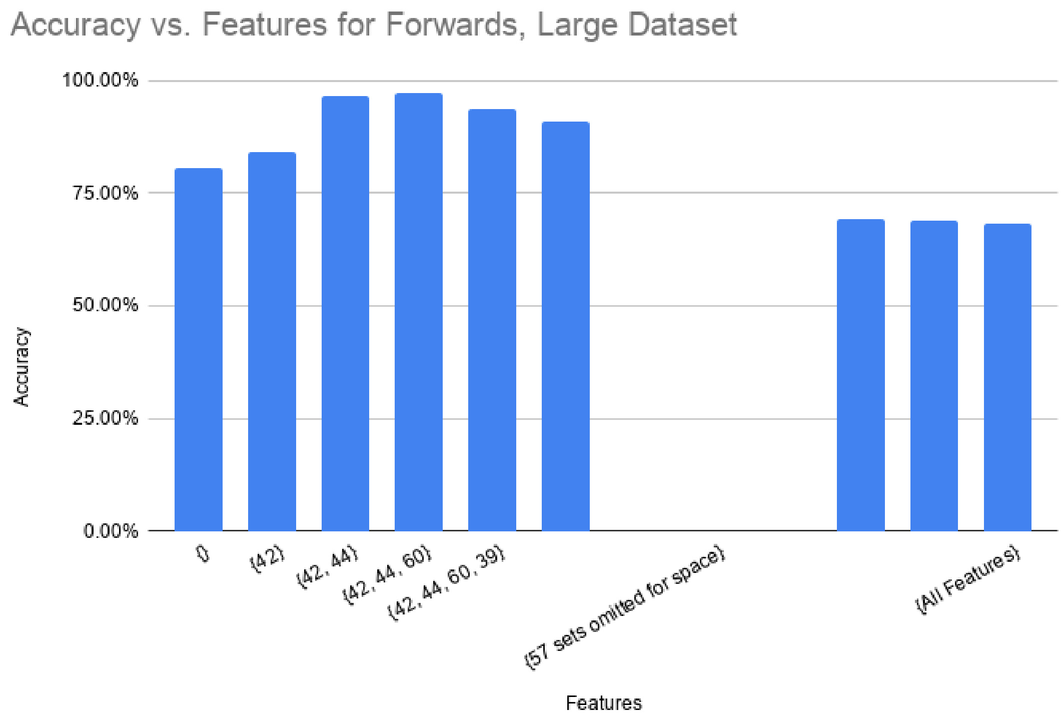

Figure 3: Accuracy of subsets of features discovered by forward selection on the large dataset. Features inbetween the 4 features and all features were omitted to save space.

As shown in Figure 3, we start with a default rate of 80.60% when no features are added for the large dataset. It increases to 84.30% accuracy when ‘42’ was added, then 96.40% when feature ‘44’ was added. It’s at its peak when feature ‘60’ is also added, resulting in an accuracy of 97.40%, {42, 44, 60}. Like the small dataset, each additional feature added after feature ‘60’ was added gradually decreased the accuracy until it reached the end with all features with an accuracy of 68.40%.

Backward Elimination on the Large Dataset

Lastly in Figure 4, we see the result of running backward elimination on the large dataset assigned to me, CS170_Large_DataSet__98.txt.

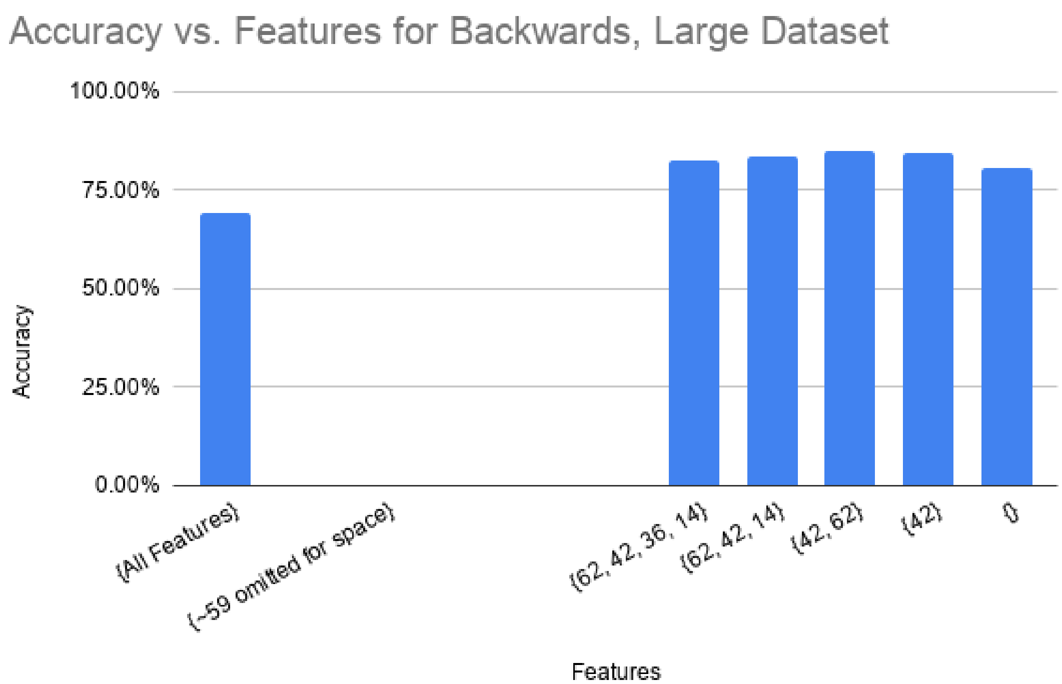

Figure 4: Accuracy of subsets of features discovered by backward elimination on the large dataset. Features between all features and 4 features were omitted to save space.

Finally, Figure 4 shows the backward elimination bar chart for large dataset. It started with all the features with an accuracy of 69.40%, which like the small dataset, gradually increased until it hit the peak at {42, 62}. Unlike the forward selection algorithm, the best feature subset for backward elimination on the large dataset is {42, 62}, which has an accuracy of 84.8%. The default rate remained the same as the forward search at 80.60% accuracy with no features.

Conclusion for the Large Dataset

In the end, Figures 3 and 4 that showcase forward selection and backward elimination on the large dataset have different best feature sets. The forward selection's best feature set was {42, 44, 60} at 97.4% accuracy, while the backward elimination's best feature was {42, 62} at 84.8% accuracy. The forward selection has a better accuracy most likely due to the fact that backward elimination will start with all 64 features, then gradually remove features, potentially removing the one(s) that would make it more accurate.

Computational effort for search:

I implemented the searches in Python3 with assistance from python libraries: NumPy and Time. I ran all experiments on 2 laptops:

- Dell Laptop, Windows OS with an AMD Ryzen 7 7730U with Radeon Graphics and 16.0 GB RAM.

- MacBook Air, MacOS 15.5 (24F74), Apple M4, 16 GB RAM

	Small Dataset (16 features, 500 instances)	Large Dataset (64 features, 3000 instances)
Forward Selection	~ 1.13 minutes	~11.9 hours
Backward Elimination	~ 1.17 minutes	~16.8 hours

Table 1: Execution times of forward selection search and backward elimination search for the small dataset and the large dataset.

Note that I ran this code before I accounted for optimizing runtime.

Traceback of the Small Data Set Forward Selection:

Welcome to the Feature Selection Algorithm
Input the name of the dataset you want to use:
Ex: CS170_Small_DataSet_85.txt

CS170_Small_DataSet__85.txt
Input the number of the algorithm you want to run
1) Forward Selection
2) Backward Selection
1
This dataset has 16 features (not including the class attribute), with 500.

Running nearest neighbor with all 16 features, using leaving-one-out evaluation, I get an accuracy of 74.6%
Beginning search...

Forward Selection Algorithm

On the 1st level of the search tree
Using feature {1}, the accuracy is 71.4%
Using feature {2}, the accuracy is 68.0%
Using feature {3}, the accuracy is 71.0%
Using feature {4}, the accuracy is 69.6%
Using feature {5}, the accuracy is 65.4%
Using feature {6}, the accuracy is 74.6%
Using feature {7}, the accuracy is 69.6%
Using feature {8}, the accuracy is 71.2%
Using feature {9}, the accuracy is 68.8%
Using feature {10}, the accuracy is 84.6%
Using feature {11}, the accuracy is 69.2%
Using feature {12}, the accuracy is 65.8%
Using feature {13}, the accuracy is 69.8%
Using feature {14}, the accuracy is 71.2%
Using feature {15}, the accuracy is 70.0%
Using feature {16}, the accuracy is 71.8%
Feature set {10} was best, accuracy is 84.6%
On the 2nd level of the search tree
Using feature {1, 10}, the accuracy is 81.0%
Using feature {10, 2}, the accuracy is 82.6%
Using feature {10, 3}, the accuracy is 79.2%
Using feature {10, 4}, the accuracy is 84.2%
Using feature {10, 5}, the accuracy is 83.8%
Using feature {10, 6}, the accuracy is 95.2%
Using feature {10, 7}, the accuracy is 80.0%
Using feature {8, 10}, the accuracy is 79.4%
Using feature {9, 10}, the accuracy is 81.8%
Using feature {10, 11}, the accuracy is 84.0%
Using feature {10, 12}, the accuracy is 85.2%
Using feature {10, 13}, the accuracy is 79.2%
Using feature {10, 14}, the accuracy is 83.4%
Using feature {10, 15}, the accuracy is 82.4%
Using feature {16, 10}, the accuracy is 83.8%
Feature set {10, 6} was best, accuracy is 95.2%
On the 3rd level of the search tree
Using feature {1, 10, 6}, the accuracy is 93.2%
Using feature {10, 2, 6}, the accuracy is 90.2%
Using feature {10, 3, 6}, the accuracy is 92.0%
Using feature {10, 4, 6}, the accuracy is 92.8%
Using feature {10, 5, 6}, the accuracy is 92.2%
Using feature {10, 6, 7}, the accuracy is 90.6%

Using feature {8, 10, 6}, the accuracy is 92.2%
Using feature {9, 10, 6}, the accuracy is 92.2%
Using feature {10, 11, 6}, the accuracy is 92.8%
Using feature {10, 12, 6}, the accuracy is 91.0%
Using feature {10, 13, 6}, the accuracy is 90.4%
Using feature {10, 6, 14}, the accuracy is 94.4%
Using feature {10, 6, 15}, the accuracy is 90.6%
Using feature {16, 10, 6}, the accuracy is 93.0%
Feature set {10, 6, 14} was best, accuracy is 94.4%
On the 4th level of the search tree
Using feature {1, 10, 6, 14}, the accuracy is 91.6%
Using feature {10, 2, 6, 14}, the accuracy is 89.4%
Using feature {10, 3, 6, 14}, the accuracy is 89.2%
Using feature {10, 4, 6, 14}, the accuracy is 90.2%
Using feature {10, 5, 6, 14}, the accuracy is 90.6%
Using feature {10, 7, 6, 14}, the accuracy is 89.6%
Using feature {8, 10, 6, 14}, the accuracy is 91.8%
Using feature {9, 10, 6, 14}, the accuracy is 90.6%
Using feature {10, 11, 6, 14}, the accuracy is 91.2%
Using feature {10, 12, 6, 14}, the accuracy is 89.4%
Using feature {10, 13, 6, 14}, the accuracy is 88.4%
Using feature {10, 15, 6, 14}, the accuracy is 90.6%
Using feature {16, 10, 6, 14}, the accuracy is 92.4%
Feature set {16, 10, 6, 14} was best, accuracy is 92.4%
On the 5th level of the search tree
Using feature {1, 6, 10, 14, 16}, the accuracy is 85.2%
Using feature {2, 6, 10, 14, 16}, the accuracy is 86.6%
Using feature {3, 6, 10, 14, 16}, the accuracy is 83.8%
Using feature {4, 6, 10, 14, 16}, the accuracy is 85.8%
Using feature {5, 6, 10, 14, 16}, the accuracy is 86.0%
Using feature {6, 7, 10, 14, 16}, the accuracy is 87.4%
Using feature {6, 8, 10, 14, 16}, the accuracy is 86.4%
Using feature {6, 9, 10, 14, 16}, the accuracy is 86.4%
Using feature {6, 10, 11, 14, 16}, the accuracy is 87.4%
Using feature {6, 10, 12, 14, 16}, the accuracy is 87.0%
Using feature {6, 10, 13, 14, 16}, the accuracy is 84.2%
Using feature {6, 10, 14, 15, 16}, the accuracy is 86.8%
Feature set {6, 7, 10, 14, 16} was best, accuracy is 87.4%
On the 6th level of the search tree
Using feature {1, 6, 7, 10, 14, 16}, the accuracy is 85.0%
Using feature {2, 6, 7, 10, 14, 16}, the accuracy is 85.2%
Using feature {3, 6, 7, 10, 14, 16}, the accuracy is 80.4%
Using feature {4, 6, 7, 10, 14, 16}, the accuracy is 84.6%
Using feature {5, 6, 7, 10, 14, 16}, the accuracy is 84.4%
Using feature {6, 7, 8, 10, 14, 16}, the accuracy is 85.0%
Using feature {6, 7, 9, 10, 14, 16}, the accuracy is 82.2%
Using feature {6, 7, 10, 11, 14, 16}, the accuracy is 82.6%
Using feature {6, 7, 10, 12, 14, 16}, the accuracy is 82.2%
Using feature {6, 7, 10, 13, 14, 16}, the accuracy is 82.0%
Using feature {6, 7, 10, 14, 15, 16}, the accuracy is 83.0%
Feature set {2, 6, 7, 10, 14, 16} was best, accuracy is 85.2%
On the 7th level of the search tree
Using feature {1, 2, 6, 7, 10, 14, 16}, the accuracy is 83.8%
Using feature {2, 3, 6, 7, 10, 14, 16}, the accuracy is 82.2%
Using feature {2, 4, 6, 7, 10, 14, 16}, the accuracy is 80.6%
Using feature {2, 5, 6, 7, 10, 14, 16}, the accuracy is 84.4%
Using feature {2, 6, 7, 8, 10, 14, 16}, the accuracy is 80.8%
Using feature {2, 6, 7, 9, 10, 14, 16}, the accuracy is 81.2%
Using feature {2, 6, 7, 10, 11, 14, 16}, the accuracy is 82.6%
Using feature {2, 6, 7, 10, 12, 14, 16}, the accuracy is 85.0%
Using feature {2, 6, 7, 10, 13, 14, 16}, the accuracy is 83.0%

Using feature {2, 6, 7, 10, 14, 15, 16}, the accuracy is 82.6%
Feature set {2, 6, 7, 10, 12, 14, 16} was best, accuracy is 85.0%
On the 8th level of the search tree
Using feature {1, 2, 6, 7, 10, 12, 14, 16}, the accuracy is 81.0%
Using feature {2, 3, 6, 7, 10, 12, 14, 16}, the accuracy is 78.8%
Using feature {2, 4, 6, 7, 10, 12, 14, 16}, the accuracy is 81.6%
Using feature {2, 5, 6, 7, 10, 12, 14, 16}, the accuracy is 80.8%
Using feature {2, 6, 7, 8, 10, 12, 14, 16}, the accuracy is 81.0%
Using feature {2, 6, 7, 9, 10, 12, 14, 16}, the accuracy is 81.4%
Using feature {2, 6, 7, 10, 11, 12, 14, 16}, the accuracy is 82.2%
Using feature {2, 6, 7, 10, 12, 13, 14, 16}, the accuracy is 81.4%
Using feature {2, 6, 7, 10, 12, 14, 15, 16}, the accuracy is 81.6%
Feature set {2, 6, 7, 10, 11, 12, 14, 16} was best, accuracy is 82.2%
On the 9th level of the search tree
Using feature {1, 2, 6, 7, 10, 11, 12, 14, 16}, the accuracy is 81.0%
Using feature {2, 3, 6, 7, 10, 11, 12, 14, 16}, the accuracy is 77.4%
Using feature {2, 4, 6, 7, 10, 11, 12, 14, 16}, the accuracy is 80.4%
Using feature {2, 5, 6, 7, 10, 11, 12, 14, 16}, the accuracy is 79.2%
Using feature {2, 6, 7, 8, 10, 11, 12, 14, 16}, the accuracy is 79.6%
Using feature {2, 6, 7, 9, 10, 11, 12, 14, 16}, the accuracy is 80.4%
Using feature {2, 6, 7, 10, 11, 12, 13, 14, 16}, the accuracy is 81.0%
Using feature {2, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 81.8%
Feature set {2, 6, 7, 10, 11, 12, 14, 15, 16} was best, accuracy is 81.8%
On the 10th level of the search tree
Using feature {1, 2, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 82.0%
Using feature {2, 3, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 77.8%
Using feature {2, 4, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 79.8%
Using feature {2, 5, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 78.4%
Using feature {2, 6, 7, 8, 10, 11, 12, 14, 15, 16}, the accuracy is 78.6%
Using feature {2, 6, 7, 9, 10, 11, 12, 14, 15, 16}, the accuracy is 78.6%
Using feature {2, 6, 7, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 79.6%
Feature set {1, 2, 6, 7, 10, 11, 12, 14, 15, 16} was best, accuracy is 82.0%
On the 11th level of the search tree
Using feature {1, 2, 3, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 78.2%
Using feature {1, 2, 4, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 79.2%
Using feature {1, 2, 5, 6, 7, 10, 11, 12, 14, 15, 16}, the accuracy is 78.8%
Using feature {1, 2, 6, 7, 8, 10, 11, 12, 14, 15, 16}, the accuracy is 83.0%
Using feature {1, 2, 6, 7, 9, 10, 11, 12, 14, 15, 16}, the accuracy is 78.4%
Using feature {1, 2, 6, 7, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.2%
Feature set {1, 2, 6, 7, 8, 10, 11, 12, 14, 15, 16} was best, accuracy is 83.0%
On the 12th level of the search tree
Using feature {1, 2, 3, 6, 7, 8, 10, 11, 12, 14, 15, 16}, the accuracy is 79.8%
Using feature {1, 2, 4, 6, 7, 8, 10, 11, 12, 14, 15, 16}, the accuracy is 80.8%
Using feature {1, 2, 5, 6, 7, 8, 10, 11, 12, 14, 15, 16}, the accuracy is 82.8%
Using feature {1, 2, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16}, the accuracy is 80.2%
Using feature {1, 2, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 80.4%
Feature set {1, 2, 5, 6, 7, 8, 10, 11, 12, 14, 15, 16} was best, accuracy is 82.8%
On the 13th level of the search tree
Using feature {1, 2, 3, 5, 6, 7, 8, 10, 11, 12, 14, 15, 16}, the accuracy is 77.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 14, 15, 16}, the accuracy is 79.2%
Using feature {1, 2, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16}, the accuracy is 76.8%
Using feature {1, 2, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 79.4%
Feature set {1, 2, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16} was best, accuracy is 79.4%
On the 14th level of the search tree
Using feature {1, 2, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.2%
Using feature {1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.2%
Using feature {1, 2, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 77.8%
Feature set {1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16} was best, accuracy is 78.2%
On the 15th level of the search tree
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.2%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.4%

Feature set {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16} was best, accuracy is 78.4%
On the 16th level of the search tree
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.6%
Feature set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16} was best, accuracy is 74.6%

Finished search!
The best feature subset is {10, 6}, which has an accuracy of 95.2%
Total time: ~ 67.96 seconds

Traceback of the Small Dataset, Backward:

Welcome to the Feature Selection Algorithm
Input the name of the dataset you want to use:
Ex: CS170_Small_DataSet__85.txt

CS170_Small_DataSet__85.txt
Input the number of the algorithm you want to run
1) Forward Selection
2) Backward Selection

2
This dataset has 16 features (not including the class attribute), with 500.

Running nearest neighbor with all 16 features, using leaving-one-out evaluation, I get an accuracy of 74.6%
Beginning search...

Backward Selection Algorithm

On the 1st level of the search tree
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.2%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 76.0%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.4%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 72.6%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 75.2%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 71.6%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 72.0%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 73.6%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.2%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 70.6%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.4%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 75.4%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.4%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.2%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 73.6%
Using feature {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 76.2%
On level 0, feature '3' was removed in current set.
Feature set {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16} was best, accuracy is 78.4%
On the 2nd level of the search tree
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 75.8%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 77.8%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.4%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 74.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 76.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 76.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.2%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 75.0%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 77.8%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.2%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 76.2%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 76.4%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 78.2%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16}, the accuracy is 79.0%
On level 1, feature '16' was removed in current set.
Feature set {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15} was best, accuracy is 79.0%
On the 3rd level of the search tree
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 75.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 82.0%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 76.2%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 78.2%

Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 77.0%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 76.8%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 77.8%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 77.0%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 77.4%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 76.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 79.6%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 75.4%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 76.4%
Using feature {1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 77.8%
On level 2, feature '2' was removed in current set.
Feature set {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15} was best, accuracy is 82.0%
On the 4th level of the search tree
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 79.8%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 79.0%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 80.0%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 78.6%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 78.2%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 78.8%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 79.8%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 77.0%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 80.6%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 81.2%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 80.2%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 78.4%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}, the accuracy is 80.8%
On level 3, feature '12' was removed in current set.
Feature set {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15} was best, accuracy is 81.2%
On the 5th level of the search tree
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 80.6%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 79.0%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 80.2%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 78.0%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 80.0%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 80.8%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 77.4%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 77.8%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 80.4%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 79.4%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 79.0%
Using feature {1, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15}, the accuracy is 78.8%
On level 4, feature '8' was removed in current set.
Feature set {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15} was best, accuracy is 80.8%
On the 6th level of the search tree
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 81.8%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 78.6%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 77.4%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 78.4%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 80.0%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 77.4%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 77.0%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 78.8%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 79.2%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 79.2%
Using feature {1, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 79.4%
On level 5, feature '1' was removed in current set.
Feature set {4, 5, 6, 7, 9, 10, 11, 13, 14, 15} was best, accuracy is 81.8%
On the 7th level of the search tree
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 79.8%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 81.2%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 78.8%

Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 81.8%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 80.2%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 75.4%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 80.0%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 80.4%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 79.2%
Using feature {4, 5, 6, 7, 9, 10, 11, 13, 14, 15}, the accuracy is 78.4%
On level 6, feature '7' was removed in current set.
Feature set {4, 5, 6, 9, 10, 11, 13, 14, 15} was best, accuracy is 81.8%
On the 8th level of the search tree
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 80.8%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 81.4%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 81.4%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 83.6%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 77.2%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 79.6%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 82.8%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 79.4%
Using feature {4, 5, 6, 9, 10, 11, 13, 14, 15}, the accuracy is 80.4%
On level 7, feature '9' was removed in current set.
Feature set {4, 5, 6, 10, 11, 13, 14, 15} was best, accuracy is 83.6%
On the 9th level of the search tree
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 83.8%
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 81.6%
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 80.0%
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 76.0%
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 82.0%
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 85.4%
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 81.2%
Using feature {4, 5, 6, 10, 11, 13, 14, 15}, the accuracy is 82.8%
On level 8, feature '13' was removed in current set.
Feature set {4, 5, 6, 10, 11, 14, 15} was best, accuracy is 85.4%
On the 10th level of the search tree
Using feature {4, 5, 6, 10, 11, 14, 15}, the accuracy is 85.6%
Using feature {4, 5, 6, 10, 11, 14, 15}, the accuracy is 83.0%
Using feature {4, 5, 6, 10, 11, 14, 15}, the accuracy is 78.2%
Using feature {4, 5, 6, 10, 11, 14, 15}, the accuracy is 76.0%
Using feature {4, 5, 6, 10, 11, 14, 15}, the accuracy is 85.8%
Using feature {4, 5, 6, 10, 11, 14, 15}, the accuracy is 84.8%
Using feature {4, 5, 6, 10, 11, 14, 15}, the accuracy is 85.8%
On level 9, feature '11' was removed in current set.
Feature set {4, 5, 6, 10, 14, 15} was best, accuracy is 85.8%
On the 11th level of the search tree
Using feature {4, 5, 6, 10, 14, 15}, the accuracy is 87.0%
Using feature {4, 5, 6, 10, 14, 15}, the accuracy is 85.6%
Using feature {4, 5, 6, 10, 14, 15}, the accuracy is 77.4%
Using feature {4, 5, 6, 10, 14, 15}, the accuracy is 76.8%
Using feature {4, 5, 6, 10, 14, 15}, the accuracy is 85.0%
Using feature {4, 5, 6, 10, 14, 15}, the accuracy is 87.8%
On level 10, feature '15' was removed in current set.
Feature set {4, 5, 6, 10, 14} was best, accuracy is 87.8%
On the 12th level of the search tree
Using feature {4, 5, 6, 10, 14}, the accuracy is 90.6%
Using feature {4, 5, 6, 10, 14}, the accuracy is 90.2%
Using feature {4, 5, 6, 10, 14}, the accuracy is 80.4%
Using feature {4, 5, 6, 10, 14}, the accuracy is 75.8%
Using feature {4, 5, 6, 10, 14}, the accuracy is 89.4%
On level 11, feature '4' was removed in current set.
Feature set {10, 5, 6, 14} was best, accuracy is 90.6%
On the 13th level of the search tree
Using feature {10, 5, 6, 14}, the accuracy is 94.4%

Using feature {10, 5, 6, 14}, the accuracy is 82.8%
Using feature {10, 5, 6, 14}, the accuracy is 76.8%
Using feature {10, 5, 6, 14}, the accuracy is 92.2%
On level 12, feature '5' was removed in current set.
Feature set {10, 6, 14} was best, accuracy is 94.4%
On the 14th level of the search tree
Using feature {10, 6, 14}, the accuracy is 83.4%
Using feature {10, 6, 14}, the accuracy is 75.0%
Using feature {10, 6, 14}, the accuracy is 95.2%
On level 13, feature '14' was removed in current set.
Feature set {10, 6} was best, accuracy is 95.2%
On the 15th level of the search tree
Using feature {10, 6}, the accuracy is 84.6%
Using feature {10, 6}, the accuracy is 74.6%
On level 14, feature '6' was removed in current set.
Feature set {10} was best, accuracy is 84.6%
On the 16th level of the search tree
(Warning: Accuracy has decreased! Continuing search in case of local maxima)

Finished search!
The best feature subset is {10, 6}, which has an accuracy of 95.2%
Total time: ~ 70.04 seconds

Link to GitHub Code Repository Here: https://github.com/kayliezhao/cs170_lab_2